The effect of the bicarbonate anion on serum ionized calcium concentration in vitro

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The effect of changes in bicarbonate ion concentration on calcium ion concentration was examined in vitro in serum and protein-free solution. The findings in this study support the formation of a calcium-bicarbonate complex ($CaHCO_3^+$) that has a K_A of 5.20 in protein-free solution. [Ca^{++}] varied inversely with [HCO_3^-] in both serum and protein-free solution. This variation was independent of the known variation of [Ca^{++}] with pH. In serum [Ca^{++}] varied 0.0036 mM Ca^{++} per 1 mM change in [HCO_3^-] compared with a variation of 0.0060 mM Ca^{++} per 0.01 unit change in pH. Addition of bicarbonate to serum (Pco_2 constant) produced a 50% greater decrease in [Ca^{++}] than that produced by a reduction in Pco_2 which gave the same pH change. These findings indicate that abnormal bicarbonate concentrations should be considered when ionized calcium concentrations are estimated from total plasma calcium values in acid-base disorders. In metabolic acid-base disorders, the bicarbonate effect adds to the pH effect on calcium ion concentration. In compensated respiratory acid-base disorders, the bicarbonate effect subtracts from the pH effect on calcium ion concentration. (J LAB CLIN MED 103:93, 1984.)

Ionized calcium is the physiologically active fraction of calcium in body fluids.[1-3] In plasma, ionized calcium constitutes 45% to 50% of total calcium; 40% to 45% of calcium in plasma is bound to plasma protein, principally albumin; the remaining 10% to 15% of calcium, or normally about 0.30 mM calcium, is complexed with the major anions in plasma—bicarbonate, phosphate, sulfate, citrate, and lactate. The steady-state ionized calcium concentration in plasma is determined by a balance between influx and efflux of calcium across intestines, kidneys, and bones. Parathyroid hormone and 1,25-dihydroxy-vitamin D regulate these processes and generally maintain plasma ionized calcium concentration within a narrow range.

The effects of protein and the anions phosphate and citrate on ionized calcium concentrations have been studied extensively[4-9] but little attention has been given to the effect of the bicarbonate anion. Bicarbonate has a low affinity for calcium compared with other anions found in the plasma; however, the concentration of bicarbonate is relatively high and recent studies suggest that the calcium-bicarbonate complex accounts for about 50% of the complexed calcium in the plasma.[10, 11] If bicarbonate normally binds 6% of total calcium in plasma, it may be an important factor affecting calcium partitioning in plasma when bicarbonate concentrations are substantially abnormal.

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The effect of a change in bicarbonate concentration on ionized calcium concentration may be estimated from the ratio of ionized, protein-bound, and complexed calcium in the plasma and the concentration of calcium bound by bicarbonate if one assumes that calcium accounted for by a change in bicarbonate concentration is divided proportionally among the three fractions (free, protein-bound, and complexed) of calcium in the plasma. According to values determined by Moore,[10] an average normal serum has 1.16 mM ionized calcium, 1.00 mM protein-bound calcium, and 0.32 mM complexed calcium. One half, or 0.16 mM, of complexed calcium may be bound by bicarbonate at a bicarbonate concentration of 25 mM.[10, 11] The molar ratio of calcium in the three fractions is approximately 7:6:2 (ionized calcium: protein-bound calcium: complexed calcium). If all bicarbonate is removed, 0.16 mM calcium would be released to be distributed among the three fractions proportionally in the ratio of 7:6:1, since the complexed calcium fraction would be reduced in half by elimination of bicarbonate. Therefore removal of 25 mM of bicarbonate should produce an increase in the ionized calcium fraction of approximately 0.08 mM Ca^{++}.

The purpose of this study was to determine whether a change in bicarbonate concentration had the effect on ionized calcium concentration that we predicted. We examined the relation between serum ionized calcium and bicarbonate concentrations in vitro, using several different methods to eliminate or control for the effect of bicarbonate concentration on serum pH and on calcium binding to protein. The experiments show that bicarbonate concentration has a major effect on ionized calcium concentration, which is independent of the effect of bicarbonate on serum pH and on calcium binding to protein.

Methods

Ionized calcium concentrations were measured with the AMT Electrion System (Applied Medical Technology, Menlo Park, Calif.).[12, 13] This system employs solid-state, calcium-selective electrodes and a glass pH electrode. The temperature of the sample is controlled by thermostat at 37° C. The pH of the sample is controlled by continuous exposure to CO_2 gas saturated with water.

Separate tanks of gas with different partial pressures of CO_2 may be used in sequence in this system to vary the pH of the sample and obtain measurements of ionized calcium at different pH values. This arrangement permits rapid determination of the relation of ionized calcium concentration to pH in a single serum sample and facilitates comparison of ionized calcium levels in sera with different bicarbonate concentrations at the same pH.

In experiment 1, serum from a single subject was divided into five 10 ml aliquots and 10.5, 21, 41, and 82 mg of $NaHCO_3$ were added to four of the aliquots. After the samples equilibrated overnight at 4° C, they were warmed to 37° C and equilibrated with 5% CO_2/95% N_2 gas.

The partial pressure of CO_2 in gaseous 5% CO_2 saturated with water vapor at a pressure of 1 atmosphere was calculated to be 35.6 mm Hg at 37° C, by using the following equation[14]: P_{CO_2} = % CO_2 (760 mm Hg − 46 mm Hg).

The bicarbonate concentrations of the aliquots of serum were calculated with the fomula: $[HCO_3^-]$ = anti-log (pH − 6.10 + log $[H_2CO_3]$), which is derived from the Henderson-Hasselbalch equation:

$$pH = 6.10 + \log \frac{[HCO_3^-]}{[H_2CO_3]}$$

$[H_2CO_3]$ was calculated from the solubility constant of CO_2, 0.0301 times the partial pressure of CO_2, 35.6 mm Hg. The pH of each solution was measured by pH meter in the AMT Electrion System. The calculated values for bicarbonate concentrations in the serum samples were 28.6 mM with no added bicarbonate and 37.1, 48.4, 68.3, and 112 mM, respectively, with increasing amounts of added bicarbonate.

The relation between ionized calcium concentration and pH in the sera with graded amounts of bicarbonate under conditions of constant P_{CO_2} was compared with the relation between ionized

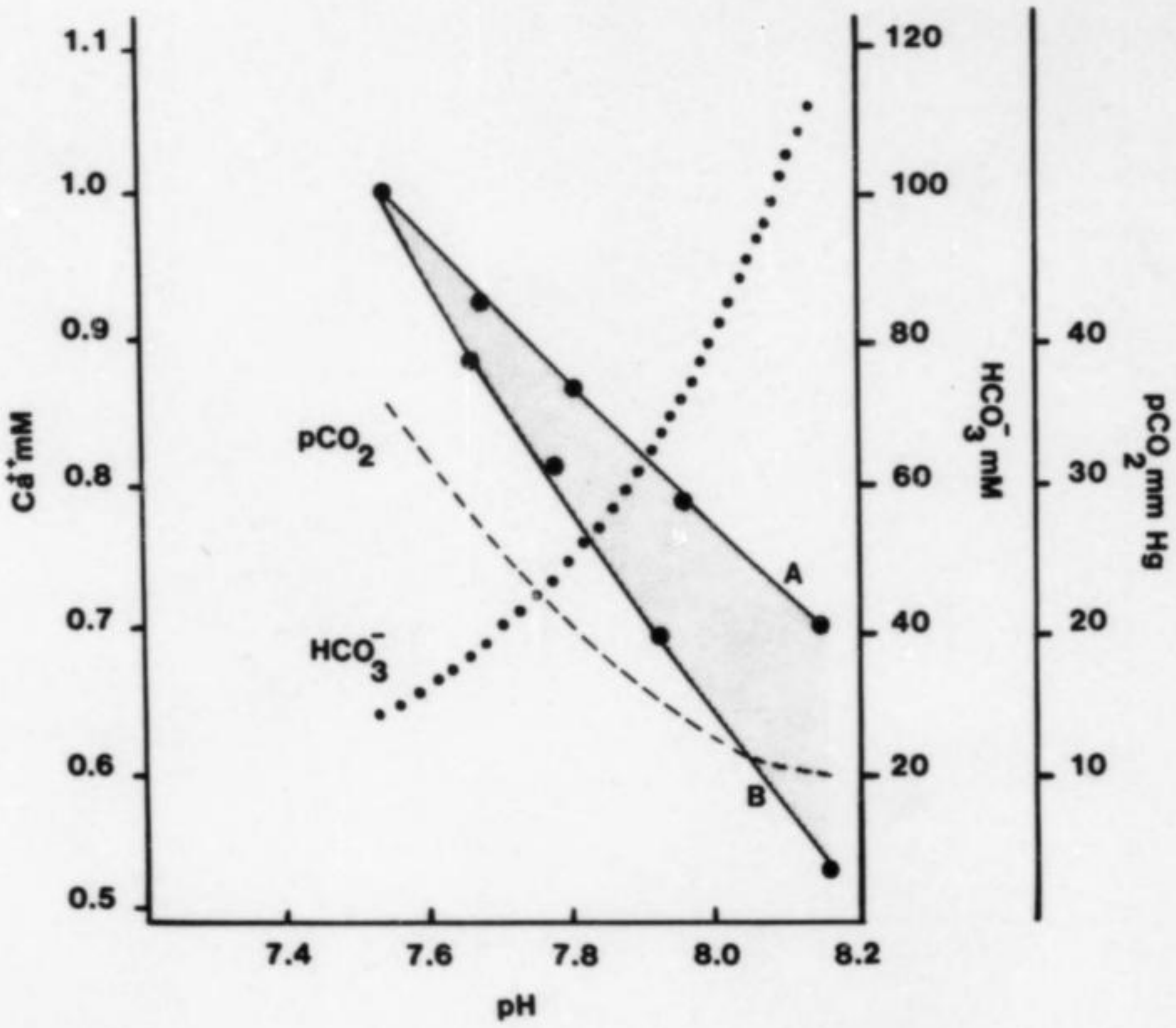

Fig. 1. Comparison of the relation between [Ca^{++}] and pH in human serum with Pco_2 variable *(curve A)* with the relation between [Ca^{++}] and pH in the same serum with Pco_2 constant (35.6 mm Hg) and HCO_3^- variable, *(curve B)*. The shaded area between the two curves represents the fall in [Ca^{++}] as a result of calcium complex formation. The dashed and dotted lines show the values for Pco_2 and HCO_3^- concentration corresponding to the pH values for *curve A* and *curve B*, respectively.

calcium and pH in serum without additional bicarbonate under conditions of decreasing Pco_2. In the latter experiment the bicarbonate concentration decreases only very slightly with a fall in Pco_2[14] and was considered to be constant for the purpose of this comparison.

In experiment 2, serum obtained from a single subject was divided into four 10 ml aliquots. Then 21 mg of $NaHCO_3$ and 5 and 10 μl of 5N HCl were added, respectively, to three of the aliquots of serum and the sera were allowed to equilibrate overnight at 4° C. On the following day the sera were warmed to 37° C and exposed sequentially to three gases: 10% CO_2/90% N_2, 5% CO_2/95% N_2, and 100% N_2. Ionized calcium values were determined under nonequilibrium conditions of both decreasing Pco_2 and increasing Pco_2. Serum HCO_3^- concentrations were measured in a Technicon AutoAnalyzer (Technicon Instruments Corp., Tarrytown, N. Y.).[15] Since the AutoAnalyzer measures free CO_2 as well as CO_2 combining power, the values determined were approximately 5% greater than the true HCO_3^- concentration.

In experiment 3, protein-free aqueous solutions were prepared containing graded concentrations of $NaHCO_3$ (10, 20, 30, 40, and 50 mM) and 1 mM of $CaCl_2$. NaCl was added to the solutions in appropriate amounts to bring the final Na concentration to 150 mM. All solutions were gassed with 100% CO_2 to establish an acid pH before $CaCl_2$ was added in order to prevent precipitation of $CaCO_3$. The concentration of ionized calcium and the relation of the concentration of ionized calcium to pH in these solutions were determined with the AMT Electrion System, under conditions of decreasing Pco_2 and increasing pH.

In experiment 4, serum from a single subject was divided into three 3 ml aliquots. Then 7 mg of $NaHCO_3$ and 3 μl of 5N HCl were added, respectively, to two of the aliquots. After the samples equilibrated overnight at 4° C they were fractionated on a 1.5 by 100 cm Bio Gel P-2 (Bio-Rad Laboratories, Richmond, Calif.) column according to the method of Toffaletti et al.[11] except that

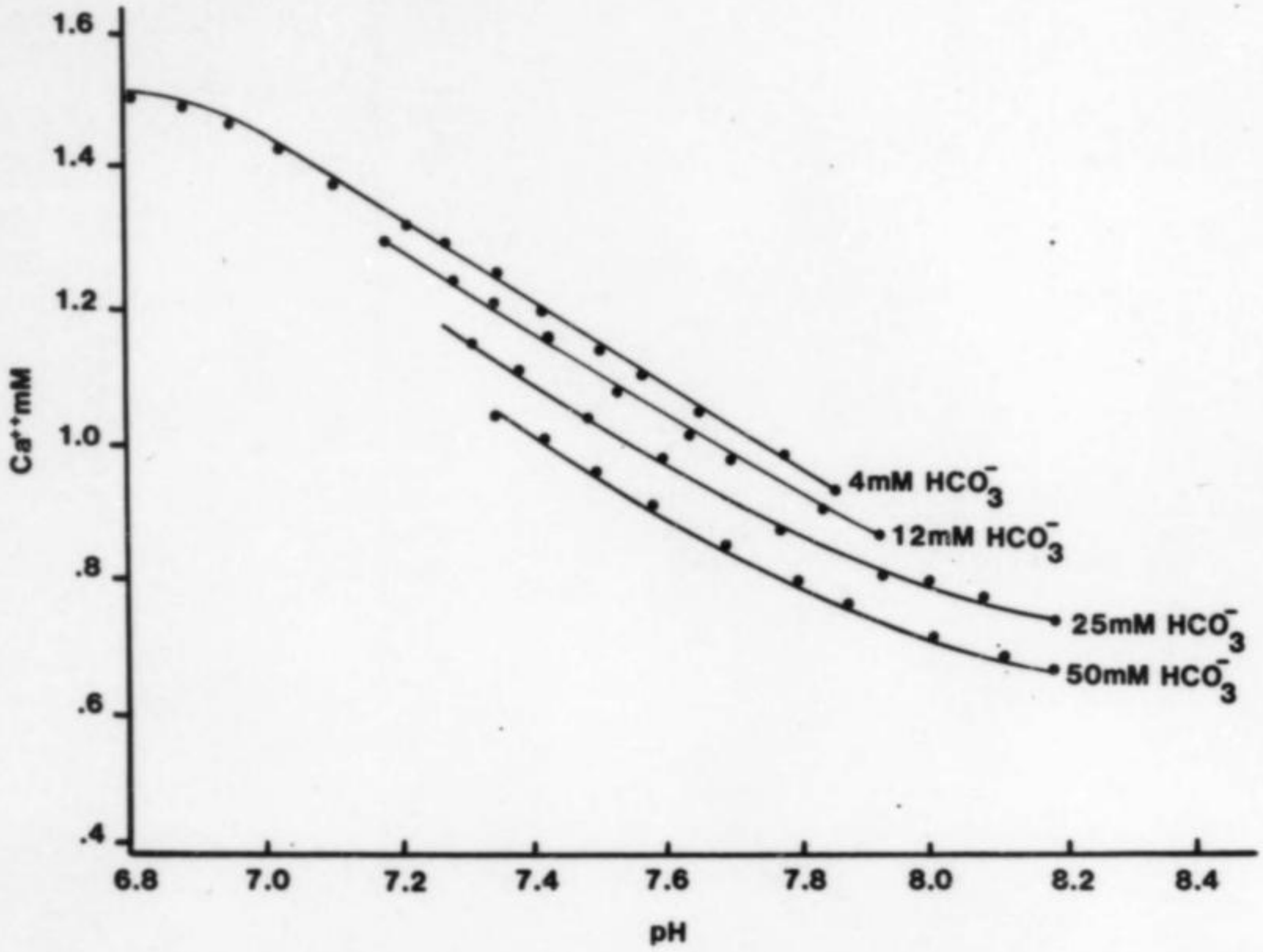

Fig. 2. Relation between $[Ca^{++}]$ and pH in separate aliquots of a single human serum containing graded concentrations of HCO_3^-. Total serum calcium concentration was 2.50 mM.

chromatography was done at 21° C. Calcium concentration in the eluted fractions was measured with a Perkin-Elmer atomic absorption spectrophotometer 303 (Perkin-Elmer Corp., Norwalk, Conn.). The elution peaks for protein and the serum anions were determined by automatic methods (total protein,[16] bicarbonate,[15] phosphate,[17] and sulfate[18]) and by manual enzymatic techniques (citrate[19] and lactate[20]).

Results

The relation of ionized calcium concentration to pH in normal serum under conditions of decreasing P_{CO_2} is shown by *curve A* in Fig. 1. The relation of ionized calcium concentration to pH in normal serum under conditions of increasing bicarbonate concentration is shown by *curve B* in Fig. 1. The fall in ionized calcium concentration resulting from a decrease in P_{CO_2} reflects increased binding of ionized calcium by protein due to alkalinization. The fall in ionized calcium concentration resulting from the addition of bicarbonate reflects both increased binding of ionized calcium by protein due to alkalinization and increased binding of ionized calcium by the additional bicarbonate. The effect of calcium complex formation alone on the ionized calcium concentration is indicated by the difference in ionized calcium values given by the two curves at the same pH. For example, when the serum bicarbonate was increased from 28.5 mM to 48.4 mM, pH rose from 7.55 to 7.78 and ionized calcium concentration fell from 1.000 mM to 0.815 mM. A 0.120 mM Ca^{++} concentration was bound to protein due to alkalinization and 0.065 mM of Ca^{++}, slightly more than one-third of the total fall of 0.185 mM of Ca^{++}, was bound by the added bicarbonate.

The effect of graded bicarbonate concentrations on serum ionized calcium under

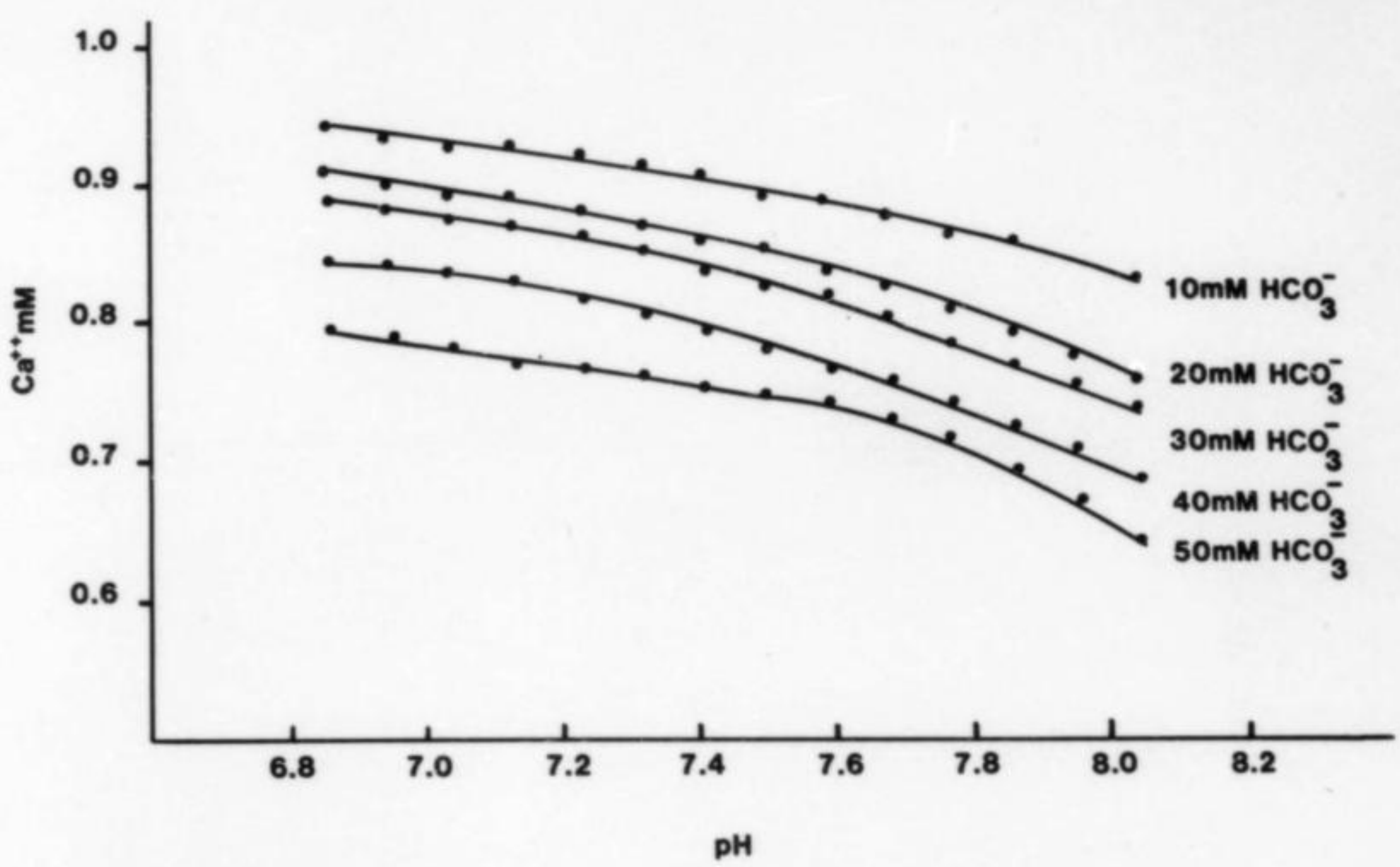

Fig. 3. Ionized calcium concentration as a function of pH in aqueous solutions of 1 mM $CaCl_2$ containing graded concentrations of HCO_3^- and 150 mM Na^+.

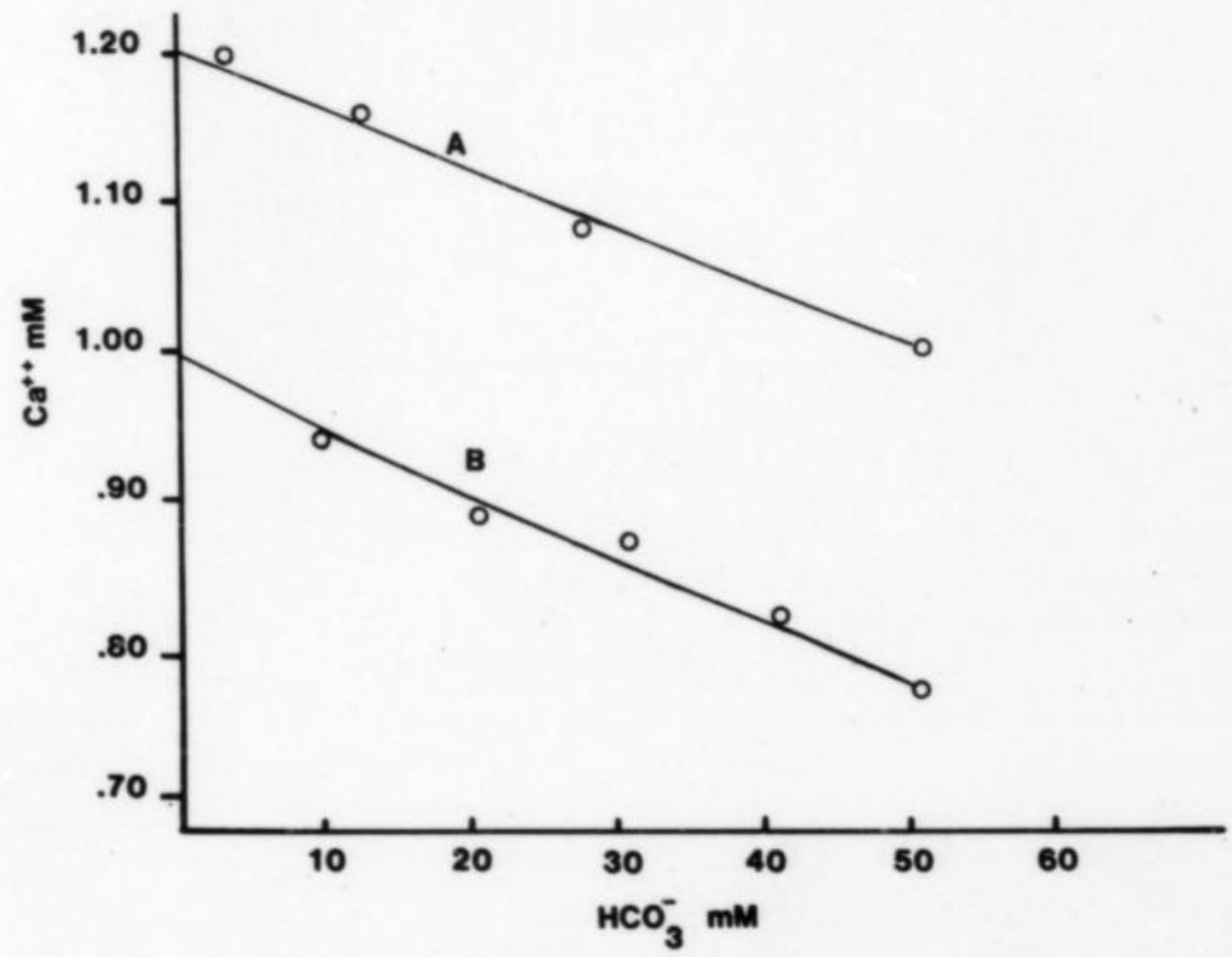

Fig. 4. Relation of Ca^{++} concentration to HCO_3^- concentration. *Line A,* Serum containing graded concentrations of HCO_3^-. Total serum calcium was 2.50 mM. Values obtained at pH 7.4 from Fig. 2. *Line B,* Protein-free solutions containing graded concentrations of HCO_3^-. Total solution calcium was 1.00 mM. Values obtained at pH 7.0 from Fig. 3. The values were obtained at pH 7.0 rather than at pH 7.4 because Ca^{++} concentration fell with an increase in pH in protein-free solution, especially above pH 7.3, and this fall is probably due to precipitation of $CaCO_3$.

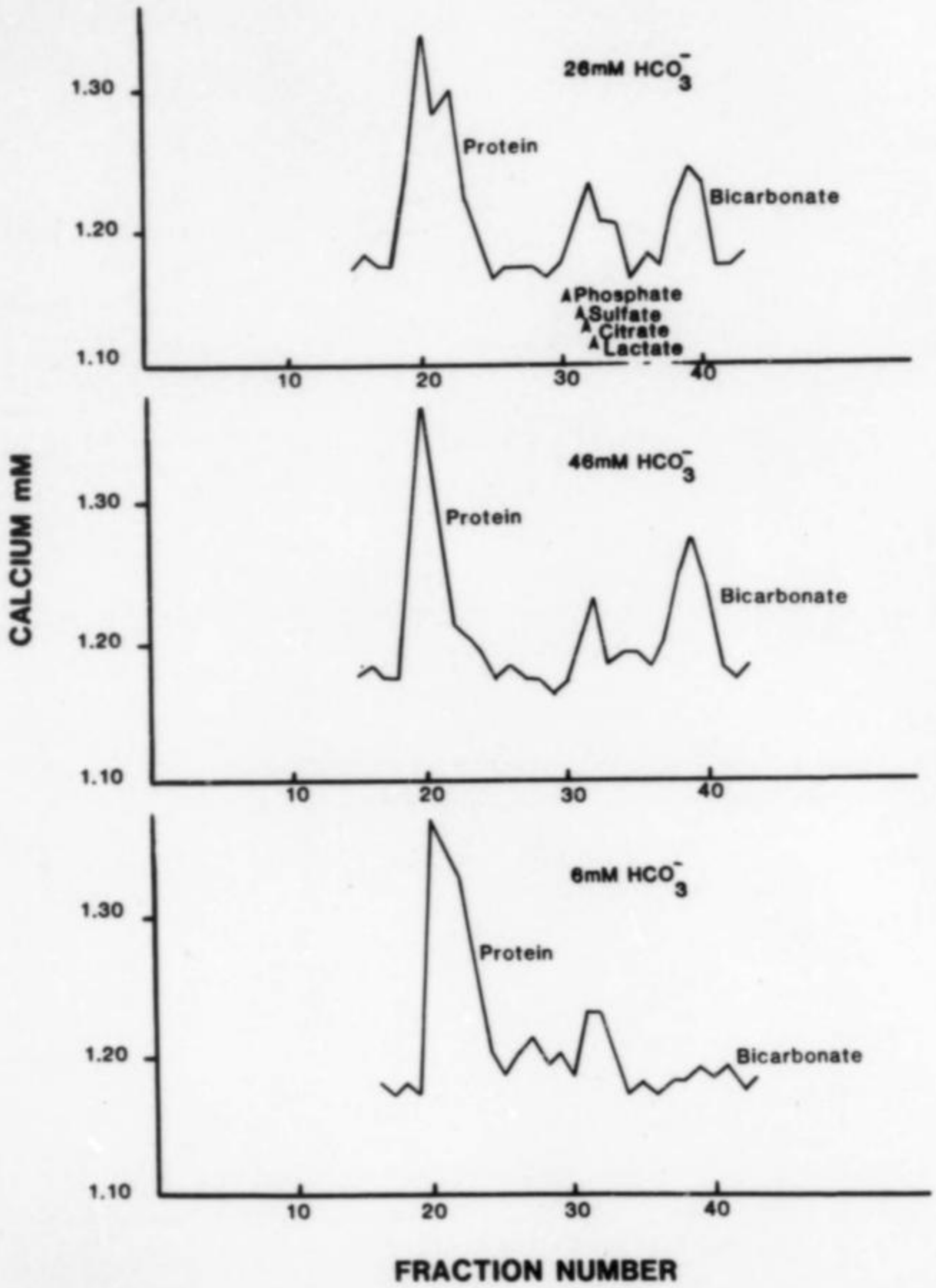

Fig. 5. Elution patterns of aliquots of a single human serum containing graded concentrations of HCO_3^- chromatographed on Bio Gel P-2. The position of the elution peaks for the anions: phosphate, sulfate, citrate, and lactate are shown in the upper panel.

conditions of varying P_{CO_2} is shown in Fig. 2. The relation between [Ca^{++}] and pH for each of the sera was reversible with variation in P_{CO_2} for pH values below 8.2. A pH difference of 0.01 resulted in an average [Ca^{++}] change of approximately 0.006 mM in pH range from 7.2 to 7.6. Schwartz[13] reported an identical value for the relation between [Ca^{++}] and pH in normal serum, using the same AMT Electrion System.

The effect of graded bicarbonate concentrations on ionized calcium concentrations in aqueous solutions of 1 mM $CaCl_2$, containing no protein, is shown in Fig. 3. Ionized calcium concentration in each solution decreased as pH increased. This pH effect on ionized calcium concentration in the absence of protein has been previously noted in ultrafiltrates of serum.[10] It may be due to formation of $CaCO_3$, since calculations show that these solutions are probably supersaturated with $CaCO_3$ above pH 6.9. The association constant (K_A) for $CaHCO_3^+$ was calculated for each of the five solutions with the following equation:

$$K_A\ CaHCO_3^+ = \frac{[Ca\ HCO_3^+]}{[Ca^{++}]\,[HCO_3^-]}$$

This equation was solved for values at pH 7.0. $CaHCO_3^+$ was assumed to be the only calcium complex formed at pH 7.0 and the concentration of $CaHCO_3^+$ was calculated as the difference between the concentration of calcium ion with no added bicarbonate (1 mM) and the measured calcium ion concentration in each solution. The mean value for K_A $CaHCO_3^+$ for the five solutions was 5.20 ± 0.35 (S.E.) L/mol.

Fig. 4 shows the relation of ionized calcium to bicarbonate concentration in serum and in protein-free solutions in experiments 2 and 3. The curve of the relation between ionized calcium concentration and bicarbonate concentration in serum shows that a change of 25 mM of bicarbonate produced a change 0.09 mM of ionized calcium, which is similar to the value of 0.08 mM of ionized calcium that we predicted from data obtained from earlier studies.[10, 11] Stated in another way, the average change in serum $[Ca^{++}]$ for a 1 mM HCO_3^- change was 0.0036 mM.

Chromatography of sera containing graded concentrations of bicarbonate showed that calcium concentration in eluted fractions containing bicarbonate was proportional to the bicarbonate concentration in the original serum (Fig. 5). The $CaHCO_3^+$ concentration and K_A $CaHCO_3^+$ were calculated from the calcium values in the fractions containing bicarbonate. In normal serum having a bicarbonate concentration of 26 mM, $CaHCO_3^+$ concentration was 0.17 mM and K_A $CaHCO_3^+$ = 5.59 L/mol. In serum with added bicarbonate having a bicarbonate concentration of 46 mM, $CaHCO_3^+$ concentration was 0.27 mM and K_A $CaHCO_3^+$ was 5.02 L/mol. In serum with added acid, having a bicarbonate concentration of 6 mM, the calcium concentration in the bicarbonate peak was not sufficiently above baseline to calculate $CaHCO_3^+$ concentration or K_A $CaHCO_3^+$.

Discussion

This study demonstrates that a change in bicarbonate concentration in serum in vitro and in protein-free solutions produces an inversely related change in ionized calcium concentration independent of pH change. Moreover, it shows that the fall in serum ionized calcium produced by an increase in bicarbonate concentration with P_{CO_2} constant is 50% greater than the fall in ionized calcium concentration produced by a decrease in P_{CO_2} that gives the same increase in pH.

The relation of ionized calcium concentration to bicarbonate concentration is most likely due to formation of a calcium-bicarbonate complex. The existence and strength of association of a calcium-bicarbonate complex has been a subject of several investigations. In 1941 Greenwald[21] studied the tendency of calcium ion to complex with the bicarbonate ion by examining the titration of bicarbonate in the presence of calcium and the solubility of $CaCO_3$ in the presence of bicarbonate. Both studies indicated that calcium formed complexes with bicarbonate. The dissociation constant K' for the $CaHCO_3^+$ complex was calculated to be 0.19 in the titration experiment and 0.15 in the solubility experiment. On the other hand, Halla and Van Tassel[22] were unable to demonstrate a calcium-bicarbonate complex in a study in which they examined the relation of pH to calcium concentration in bicarbonate solutions. However, the technique used in this study was probably too insensitive to detect a calcium-bicarbonate complex.[23] Neuman et al.[24] used an ion exchange resin method to investigate the possibility of formation of calcium-bicarbonate complexes.

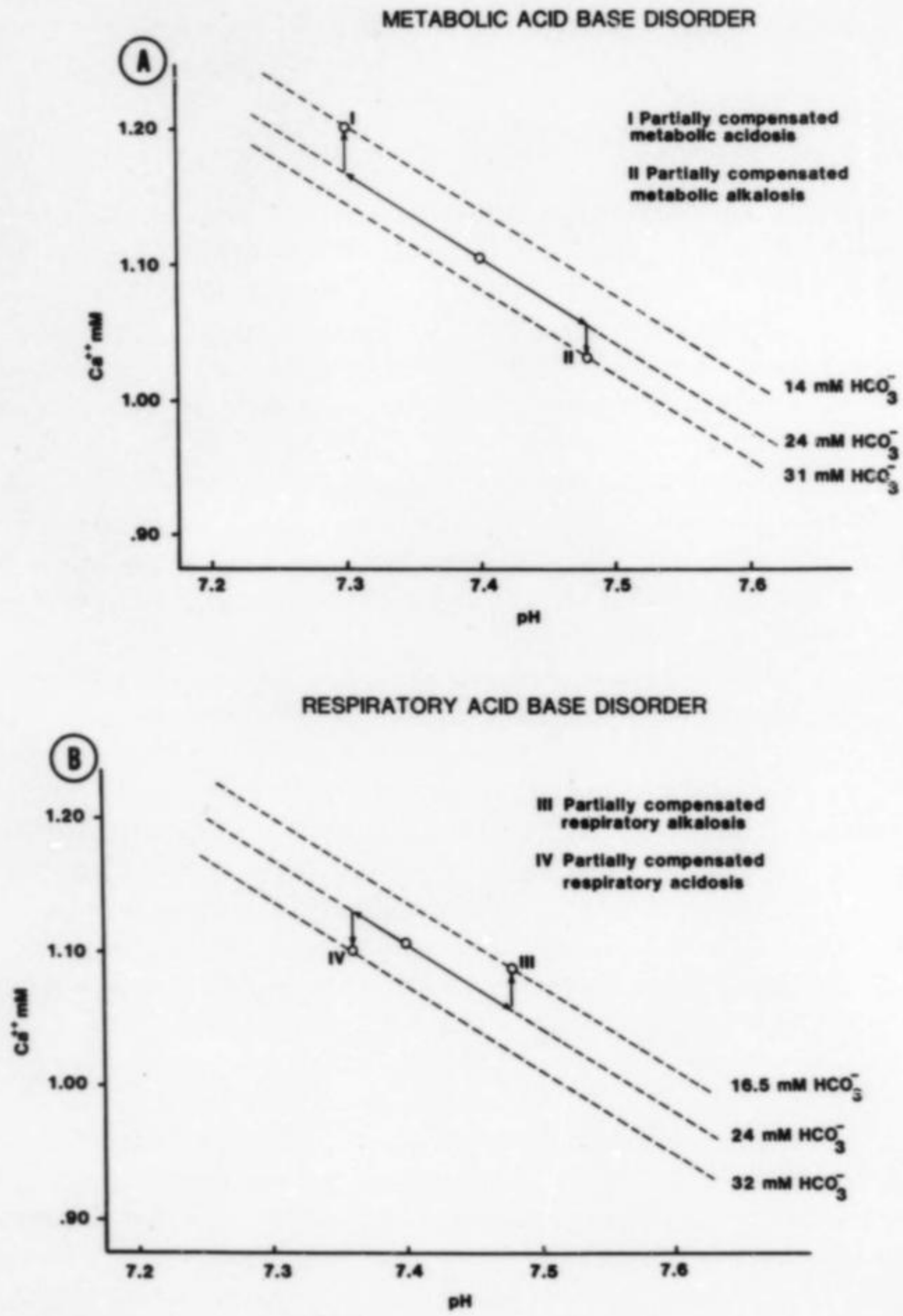

Fig. 6. Examples of changes in ionized calcium values from a normal value of 1.107 mM at P_{CO_2} 40 mm Hg, $[HCO_3^-]$ 24 mM, and pH 7.4 with alterations in $[HCO_3^-]$ and pH in conditions of acid-base disorder. The values for P_{CO_2}, $[HCO_3^-]$ and pH in these clinical examples were obtained from Davenport.[14] *Dashed lines* show the ionized calcium-pH relationship for each bicarbonate concentration in the clinical examples. *Circles* show calculated ionized calcium concentration and pH of the clinical examples. *Arrows* show the effects of change in pH and bicarbonate concentration on the variation in ionized calcium concentration from the normal value.

	P_{CO_2} *(mm Hg)*	$[HCO_3^-]$ *mM*	*pH*	$[Ca^{++}]$
Normal	40	24	7.40	1.107
I	29	14	7.30	1.203
II	43	31	7.48	1.034
III	23	16.5	7.48	1.086
IV	58	32	7.36	1.102

Their study supported the existence of a $CaHCO_3^+$ complex but determined that is dissociation constant was 0.41. These experimenters concluded that the calcium-bicarbonate complex was of doubtful physiologic significance because it was so highly dissociated. More recent studies,[10, 11, 25] including the present study, using calcium ion electrode and gel-filtration techniques strongly support the existence of a calcium-bicarbonate complex and indicate that its dissociation constant is similar to the value ($K' = 0.19$ or $K_A = 5.26$) originally determined by Greenwald[21] by titration of bicarbonate in the presence of calcium ion.

In clinical practice ionized calcium values are usually estimated from the total serum calcium concentration, with consideration of the effects of any abnormalities in serum albumin concentration and pH. The findings in this study suggest that abnormal bicarbonate concentrations should be considered as well. Abnormal bicarbonate concentrations are found in metabolic acidosis and metabolic alkalosis and in the presence of renal compensation for respiratory acidosis and respiratory alkalosis.

In metabolic acid-base disturbances, the effects on ionized calcium concentration of abnormal bicarbonate concentrations and pH are in the same direction and are additive, resulting in a greater change in ionized calcium values than the pH change alone produces (Fig. 6, *A*). In compensated respiratory acid-base disturbances, the effects of abnormal bicarbonate concentrations and pH are in opposite directions, resulting in a smaller change in ionized calcium values than the pH change alone produces (Fig. 6, *B*). In both metabolic and respiratory acid-base disturbances, compensating mechanisms reduce the change in pH and, in turn, the effect of pH change on protein-calcium binding and ionized calcium values. Therefore, in compensated acidosis or alkalosis, the effect of changes in bicarbonate concentration on ionized calcium values in some instances may be equal to or even greater than the effect of changes in pH.

The relative effects of changes in bicarbonate concentration and pH on serum ionized calcium concentrations in various clinical acid-base disorders may be calculated from our in vitro data which show that ionized calcium concentration changes approximately 0.006 mM for each 0.01 U change in pH and 0.0036 mM for each 1 mM change in HCO_3^- concentration. For example (Fig. 6, *B*) comparison of values for $[HCO_3^-]$ and pH in a case of compensated respiratory acidosis (Pco_2 58 mm Hg, HCO_3^- 32 mM, pH 7.36) with normal values (Pco_2 40 mm Hg, HCO_3^- 24 mM, pH 7.40) shows that bicarbonate concentration is increased by 8 mM and pH is decreased by 0.04 pH units. In this case, ionized calcium is increased $4 \times 0.006 = 0.024$ mM by the fall in pH but it is decreased $8 \times 0.0036 = 0.0288$ mM by the rise in bicarbonate concentration, producing a net decrease in ionized calcium of 0.0048 mM. It should be noted that actual changes in plasma ionized calcium concentrations in acid-base disturbances cannot be predicted in this way, since factors in addition to bicarbonate concentration and pH such as parathyroid hormone and 1,25-dihydroxyvitamin D concentrations may be expected to change and affect the final ionized calcium concentration in the body.

Changes in plasma bicarbonate concentration may play a role in calcium physiology in clinical situations such as the following: (1) the decrease in ionized calcium concentration associated with the postprandial alkaline tide, (2) the increase in ionized calcium concentration in primary hyperparathyroidism, and (3) the increase in total calcium concentration associated with the use of thiazide diuretics.

In a previous study in normal subjects, stimulation of gastric acid secretion and the alkaline tide was associated with a 5% fall in serum ionized calcium concentration.[26] Increased binding of calcium by protein due to a rise in the pH of the plasma can explain only about two thirds of this fall. The finding in the present study supports the hypothesis that a substantial portion of a fall in postprandial ionized calcium values is due to the increase in plasma bicarbonate concentration and $CaHCO_3^+$ complex formation.

In primary hyperparathyroidism, parathyroid hormone inhibits the reabsorption of bicarbonate by the proximal tubule in the kidney, producing a significant fall in plasma bicarbonate concentration in some patients. The fall in bicarbonate concentration in these patients may increase the ratio of ionized calcium to total calcium in the plasma.

Calcium-bicarbonate complex formation may play a role in the hypercalcemia associated with the use of thiazide diuretics. Thiazides produce an increase in plasma bicarbonate concentration. The rise in plasma bicarbonate concentration may increase total plasma calcium concentration by raising the amount of calcium bound in complexes. The effect of thiazide on the concentration of the $CaHCO_3^+$ complex in the plasma has not been specifically measured but one report indicates that the total concentration of calcium complexes is increased in patients taking thiazide.[27]

In summary this study shows that a change in the bicarbonate anion concentration has an important direct effect on calcium partitioning in the blood independent of its indirect effect through pH change. When ionized calcium values are estimated from total serum calcium concentrations in clinical situations, the effect of an abnormal bicarbonate concentration should be considered. The possible role of the bicarbonate anion in various clinical disturbances in calcium metabolism is speculative but warrants future investigation.

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